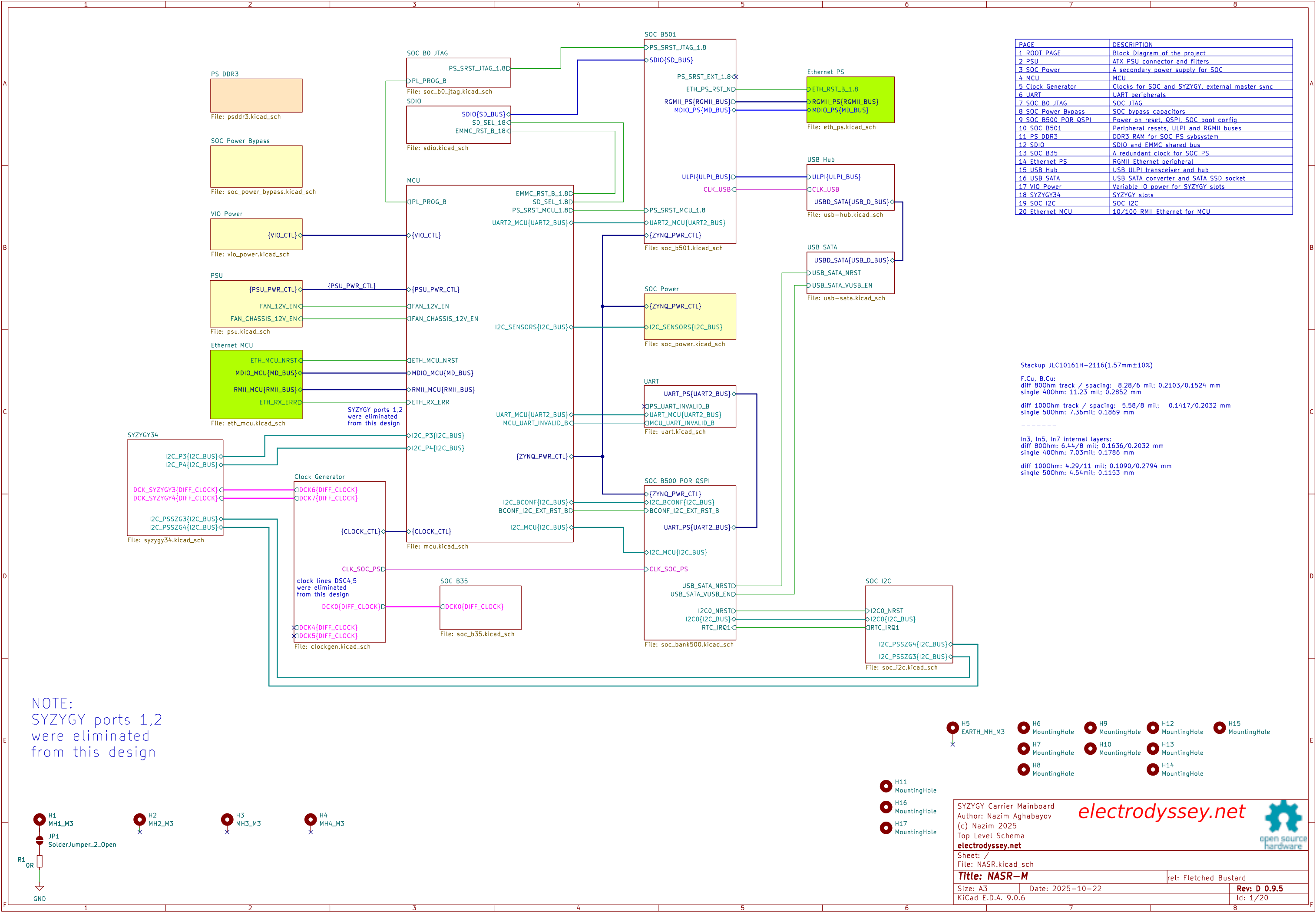
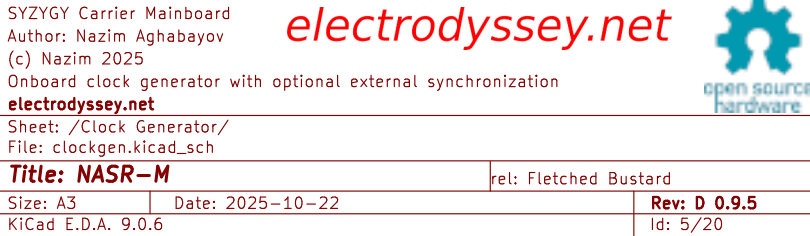
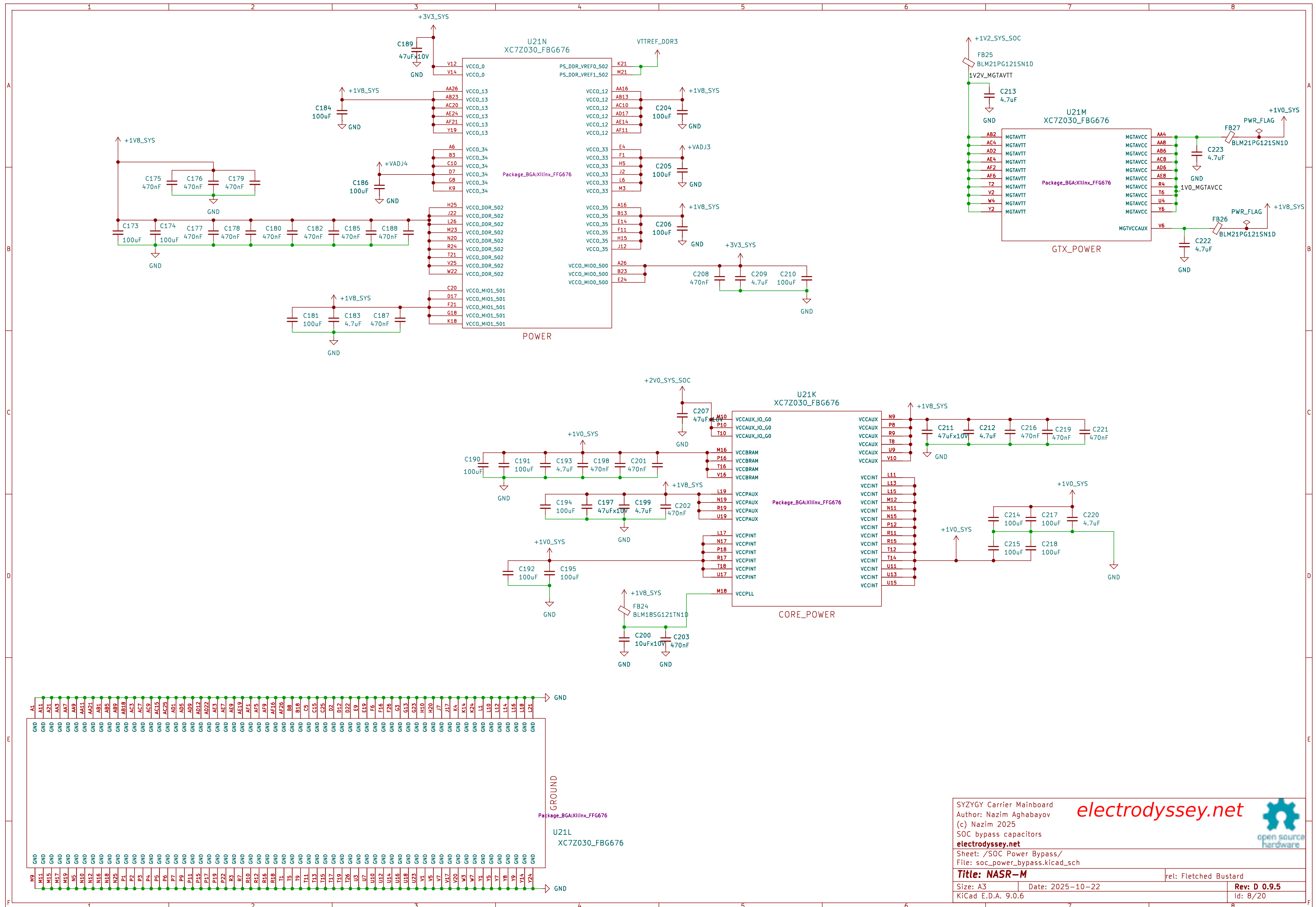








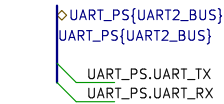
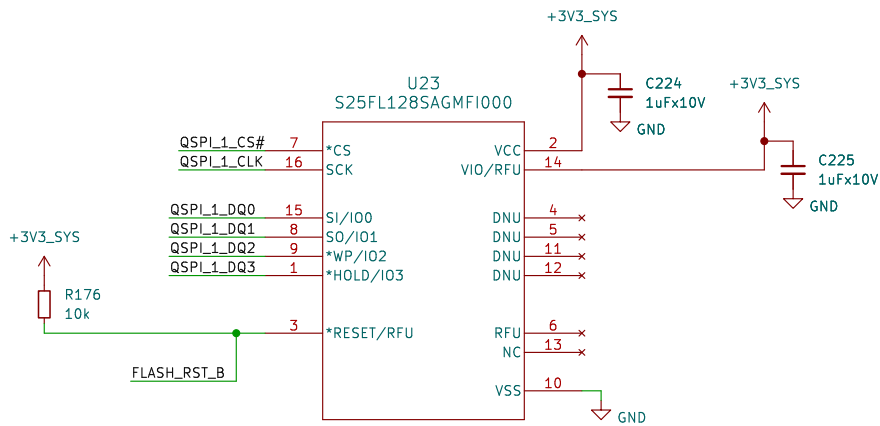
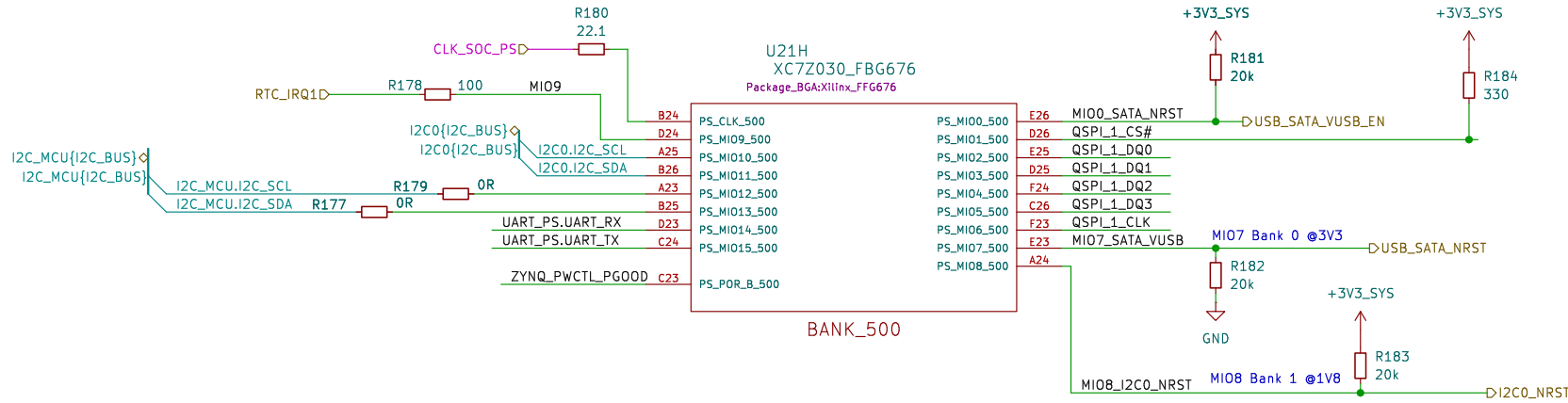
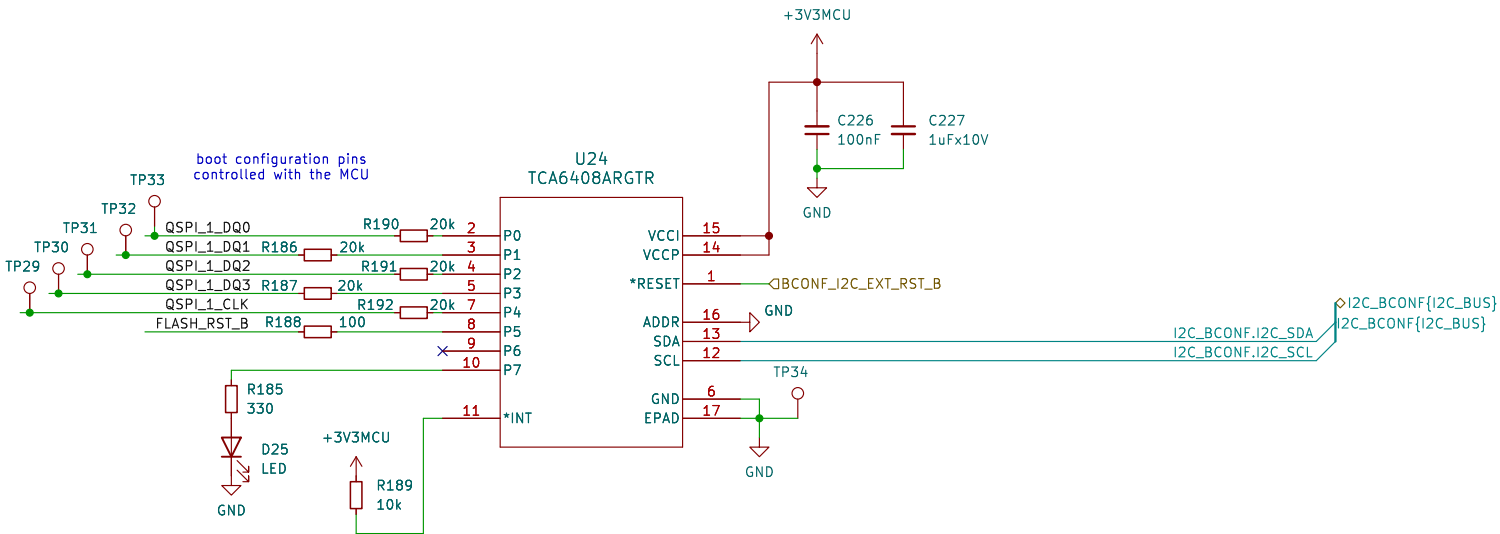
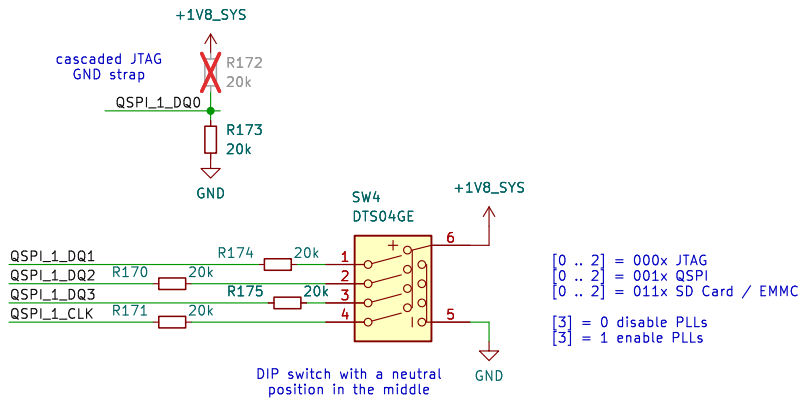
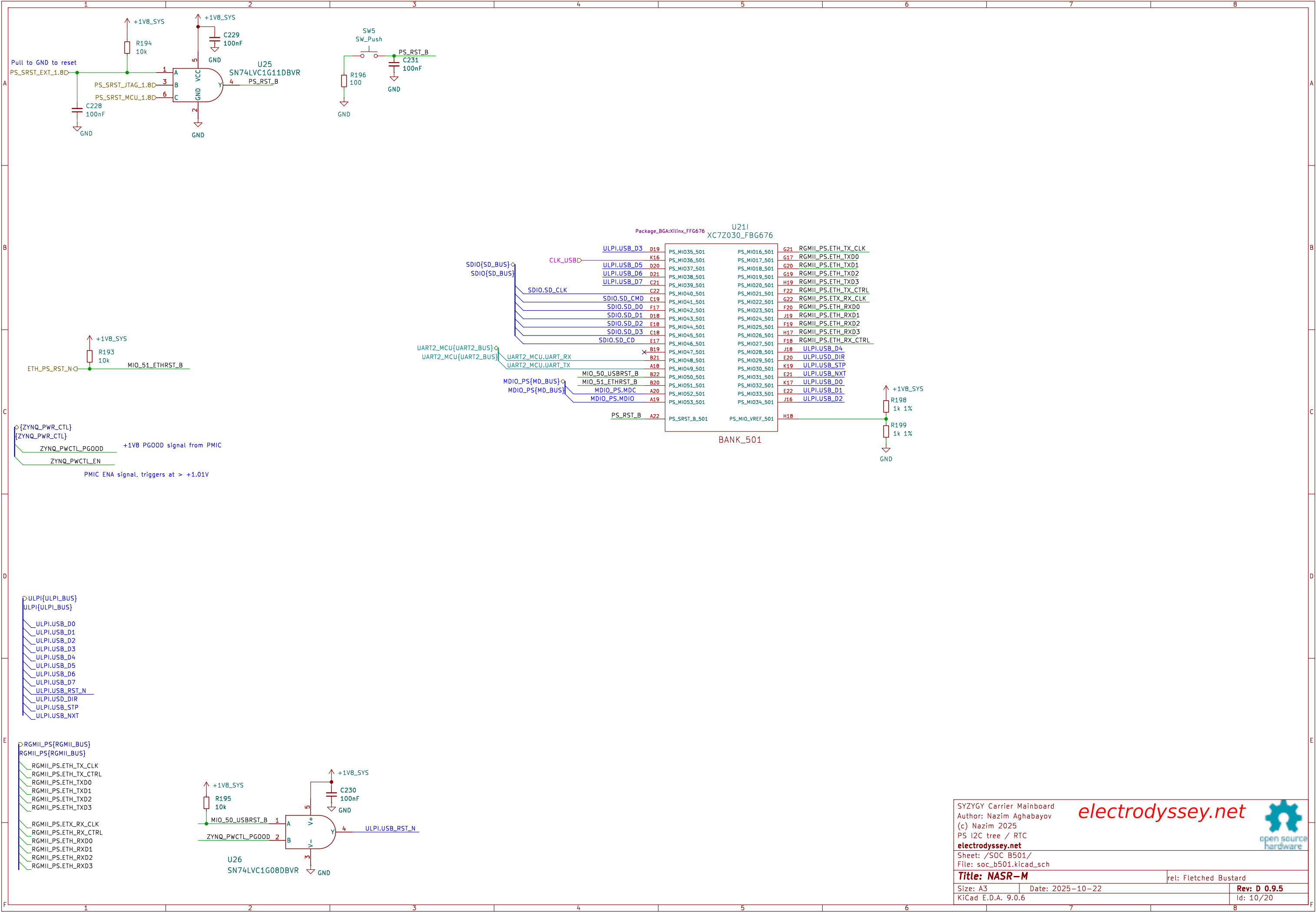


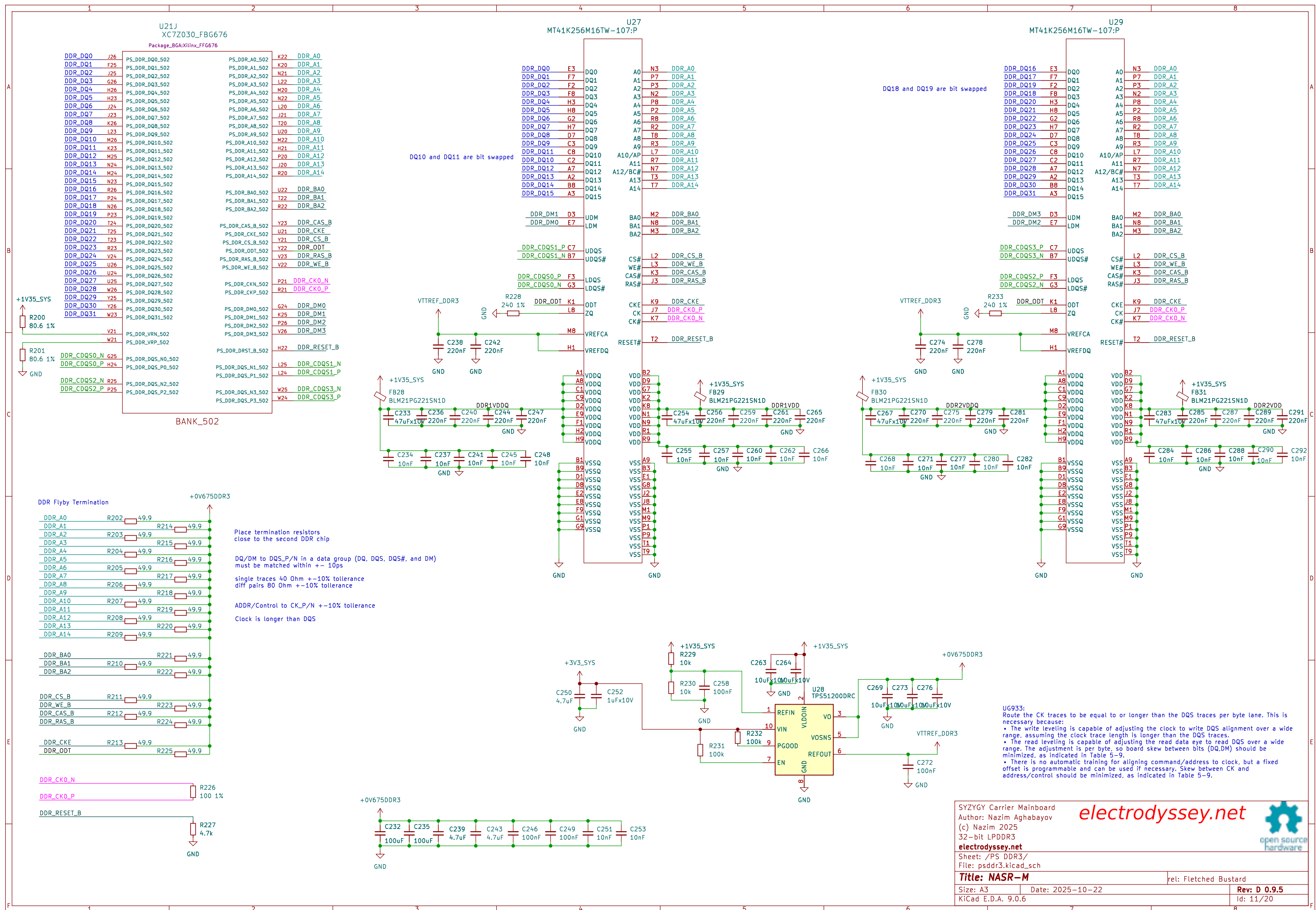
Table 6-4: Boot Mode MIO Strapping Pins

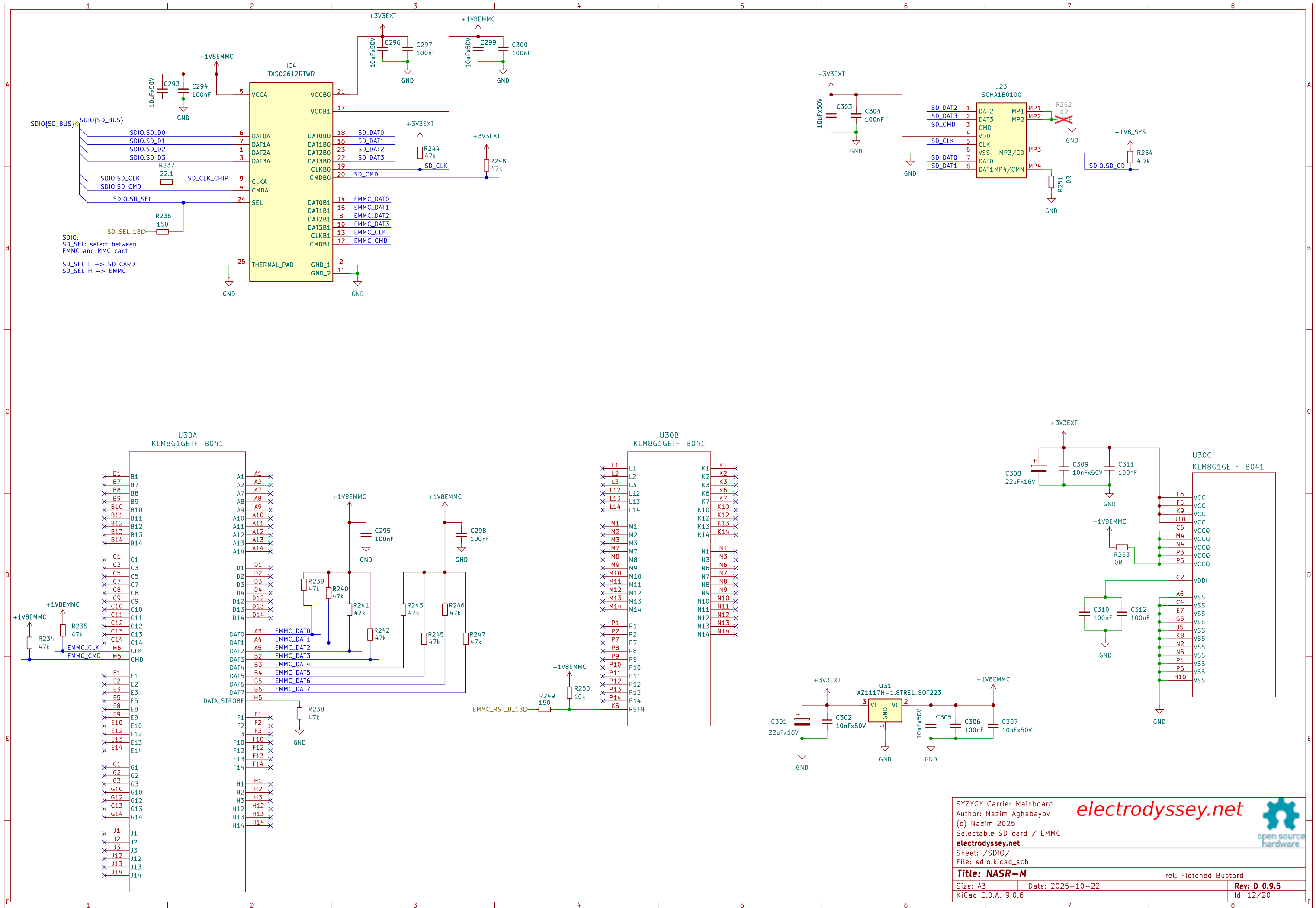
Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾			0	0	0	JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode	
NOR Boot ⁽³⁾			0	0	1		
NAND			0	1	0		
Quad-SPI ⁽³⁾			1	0	0		
SD Card			1	1	0		
Mode for all 3 PLLs							
PLL Enabled		0	Hardware waits for PLL to lock, then executes BootROM.				
PLL Bypassed		1	Allows for a wide PS_CLK frequency range.				
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

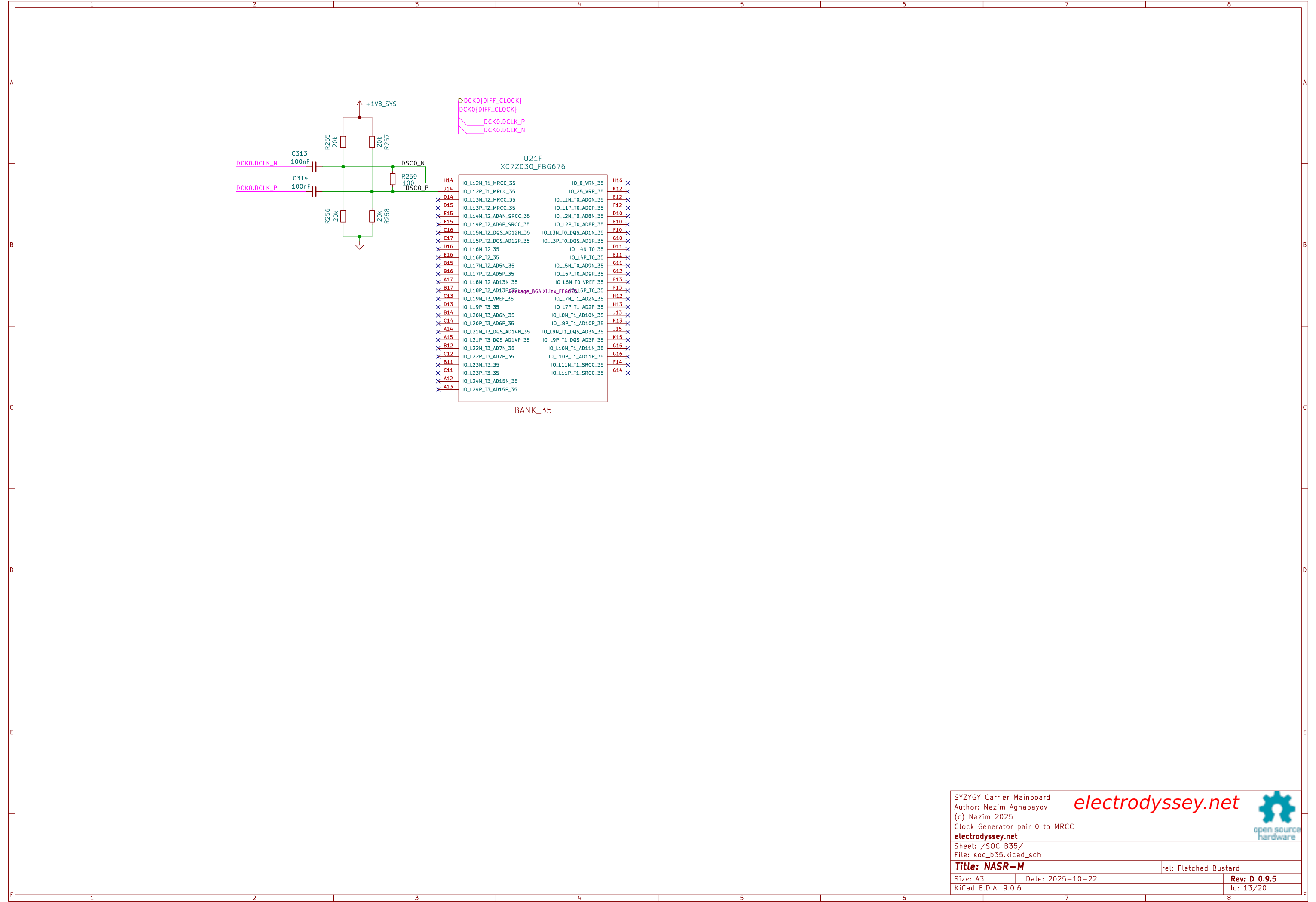
- Notes:
- JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
 - For secure mode, JTAG is not enabled and MIO[2] is ignored.
 - The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure).
 - Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

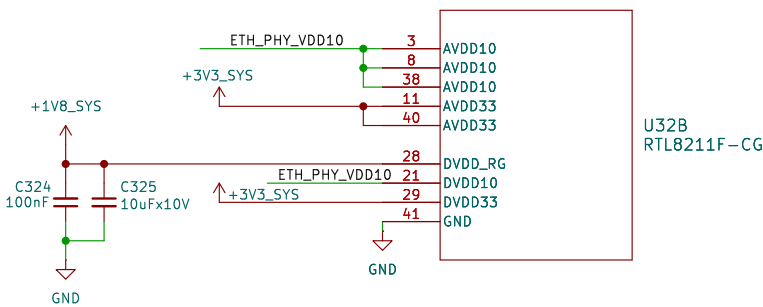
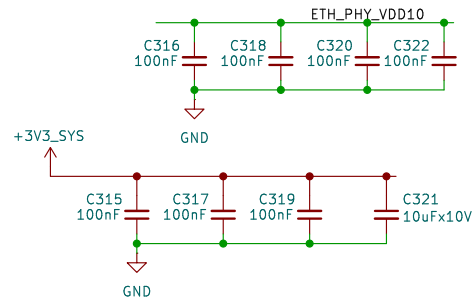
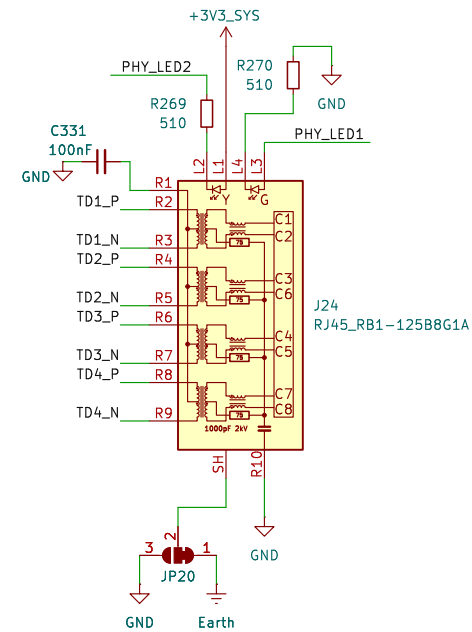
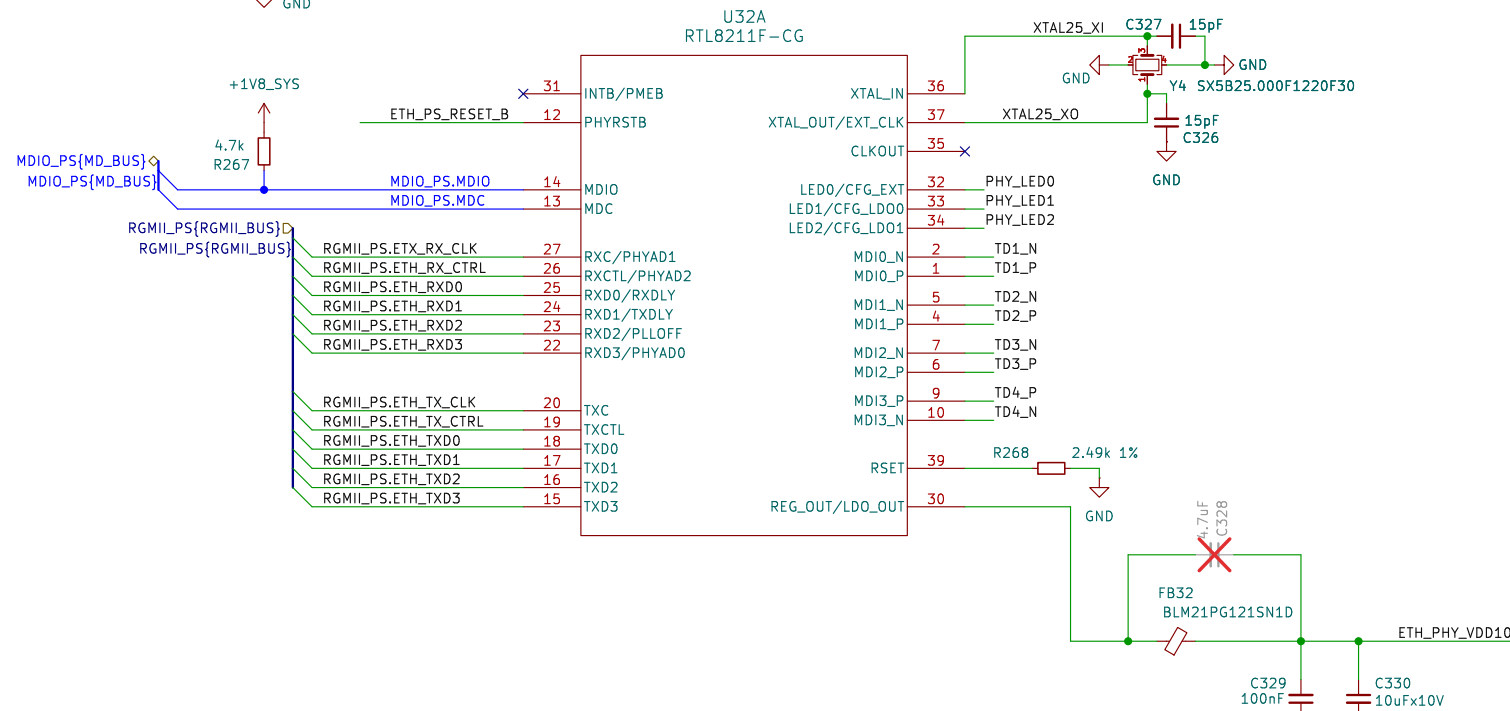
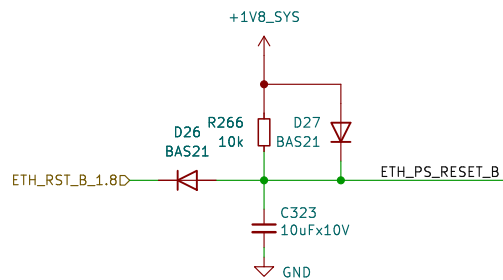










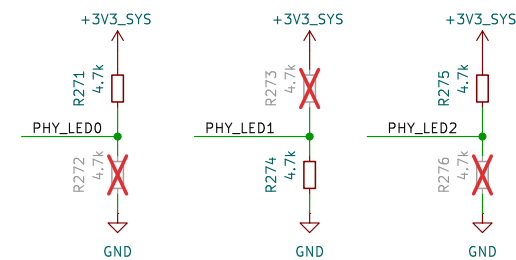


CONFIG Pin	Configuration
RXD3	PHYAD[0]
RXC	PHYAD[1]
RXCTL	PHYAD[2]
RXD2	PLLOFF
RXD1	TXDLY
RXD0	RXDLY
LED0	CFG_EXT
LED1	CFG_LDO[0]
LED2	CFG_LDO[1]

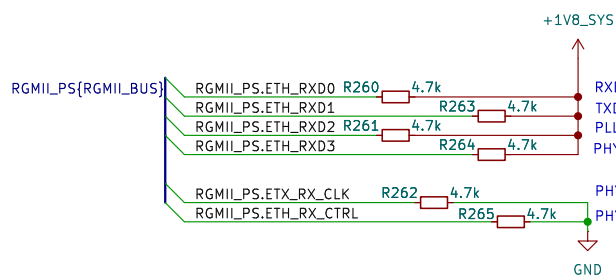
Configuration	Description
PHYAD[2:0]	PHY Address. PHYAD sets the PHY address for the device. The RTL8211F(1)/RTL8211FD(1) supports PHY addresses from 00001 to 00111. <i>Note 1: An MDIO command with PHY address=0 is a broadcast from the MAC; each PHY device should respond. This function can be disabled by setting Reg24.13=0 (See Table 37).</i> <i>Note 2: The RTL8211F(1)/RTL8211FD(1) with PHYAD[2:0]=000 can automatically remember the first non-zero PHY address. This function can be enabled by setting Reg24.6=1 (See Table 37).</i>
PLLOFF	ALDPS Mode PLL Off Configuration. 1: Stop PLL when entering ALDPS mode (via 4.7k-ohm to DVDD_RG) 0: PLL continue toggling when entering ALDPS mode (via 4.7k-ohm to GND)
TXDLY	RGMII Transmit Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
RXDLY	RGMII Receive Clock Timing Control. 1: Add 2ns delay to RXC for RXD latching (via 4.7k-ohm to DVDD_RG) 0: No delay (via 4.7k-ohm to GND)
CFG_EXT	I/O Pad External Power Source Mode Configuration. 1: Use the external power source for the IO pad (via 4.7k-ohm to 3.3V) 0: Use the integrated LDO to transform the desired voltage for the IO pad (via 4.7k-ohm to GND)

Configuration straps:

LED0 CFG_EXT
LED1 CFG_LDO0
LED2 CFG_LDO1



32	CFG_EXT	O/LI/PD	IO Pad External Power Source Mode Configuration. Pull up to use the external power source for the IO pad. Pull down to use the integrated LDO to transform the desired voltage for the IO pad.
33	CFG_LDO0	O/LI/PU	LDO Output Voltage Selection for I/O Pad/ External Power Source Voltage Selection for I/O Pad.
34	CFG_LDO1	O/LI/PD	When pulling down CFG_EXT pin, CFG_LDO[1:0] represent LDO output voltage setting for IO pad: 2'b00: Reserved. 2'b01: 2.5V. 2'b10: 1.8V. 2'b11: 1.5V. When pulling up CFG_EXT pin, CFG_LDO[1:0] stand for input voltage selection of the external power for IO pad: 2'b00: 3.3V. 2'b01: 2.5V. 2'b10: 1.8V. 2'b11: 1.5V.



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Sheet: /Ethernet_Ps/
File: eth_ps.kicad_sch

Title: NASR-M

rel: Fletched Bustard

Size: A3

Date: 2025-10-22

Rev: D 0.9.5

KiCad E.D.A. 9.0.6

Id: 14/20

